

2D Arrays (Matrix → Matrices)

- Sequential storage of elements
- Store similar type of data
- Both length and breadth

int mat[][] = new int [3][4]

↓
data type
of 2D array

↓
name of
2D array

row

	0	1	2	3
0	0,0	0,1	0,2	0,3
1	1,0	1,1	1,2	1,3
2	2,0	2,1	2,2	2,3

col

Mat [3][4]

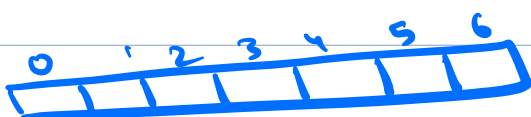
Total
cells = 12



① Access el. → mat [row-id][col-id]

Eg. mat [1][2]

int ar[] = new int[7]



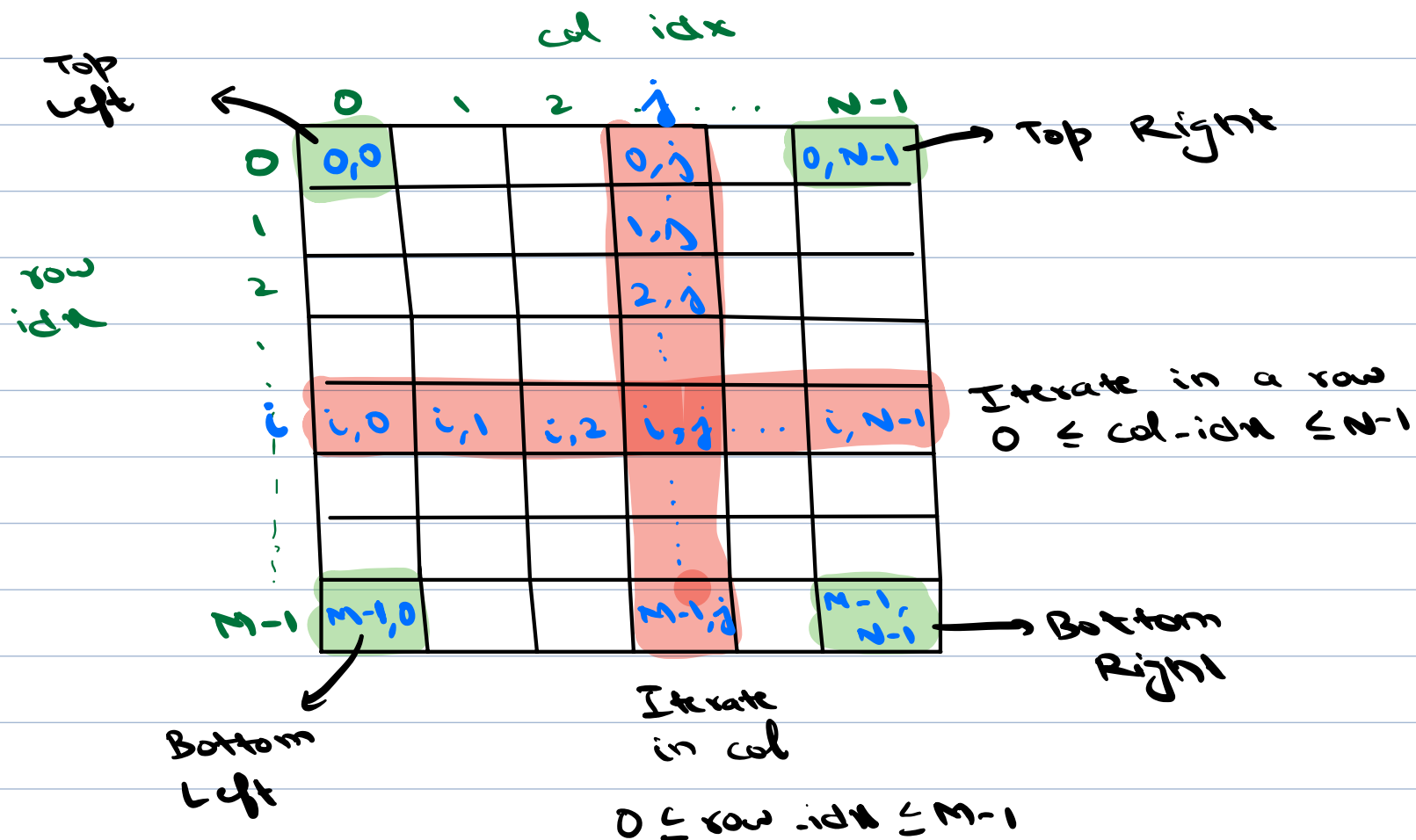
`int mat[ ][ ] = new int [5][2]`

Create a matrix of 5 columns and 7 rows !

`int mat[ ][ ] = new int [7][5]` rows cols

Mat [M] [N]

M rows, N cols
Total ele = $M \times N$



mat 3x4

	0	1	2	3
r ₀	6	8	2	9
r ₁	7	5	20	15
r ₂	1	10	14	19

[[6, 8, 2, 9],
 [7, 5, 20, 15],
 [1, 10, 14, 19]]



2D Array →

Array of arrays

Array of rows

[r₀,
 r₁,
 r₂]

mat.length → no. of rows // 3

mat[0].length → no. of cols // 4



r₀.length

3 id → 2

5

2 3 1 4 2

✓ 3 (pos)

5 (ele)

0	1	2	3	4	5
2	3	1	4	2	-

```
int n = scn.nextInt();
int arr[] = new int[n+1];

for (int i = 0 ; i < n ; i++) {
    arr[i] = scn.nextInt();
}
```

```
int pos = scn.nextInt();
int ele = scn.nextInt();
pos--
```

```
for (int i = N-1 ; i >= pos ; i--) {
    arr[i+1] = arr[i];
}
```

```
arr[pos] = ele
```

Q. $N \rightarrow$

$$1 \leq N \leq 100$$

$$1 \leq M \leq 10^3$$

`print ("Ram" + "Shyam")` Ramshyam

`print ("My age is :" + 25)` My age is 25

`print ("My age is :" + 25 + 10)`
str + int

"My age is: 25" + 10
str + int

My age is: 25 10

`print (25 + 10 + " is my age")`
35 + " is my age")

35 is my age

$N = 2$

*
**

$N = 3$

*
**

| - - *
| - * *
| * * *

| *
| * - *
| * - * - *

spaces = N - 1

for (line = 1; line ≤ N; line++) {

for (cnt = 1; cnt ≤ spaces; cnt++)
print(" ")

for (cnt = 1; cnt ≤ line; cnt++)
print("* ")

spaces--

System.out.println()

}

| - - - *
| - - * - *
| - * - * - *
| * - * - * - *

0	1	2	3	4
x	y	z	a	b

i, j

$N = 5$

i

	0	1	2	3	4
0	0,0	0,1	0,2	0,3	0,4
1	1,0	1,1	1,2	1,3	1,4
2	2,0	2,1	2,2	2,3	2,4
3	3,0	3,1	3,2	3,3	3,4
4	4,0	4,1	4,2	4,3	4,4

$i=0 \quad j=1 \text{ to } 4$

$i=1 \quad j=2 \text{ to } 4$

$i=2 \quad j=3 \text{ to } 4$

$i=3 \quad j=4 \text{ to } 4$

$i=4$

$i \quad j$
 $i+1$

Array List
↓
5, 7, 8